

Rudi: Calculadora Rudimentar

Projetando um caminho de dados

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**INSTITUTO
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Santa Catarina

Câmpus
São José

Objetivo

- Construir circuitos utilizando blocos de construção digital.
- Introdução ao caminho de dados e à unidade de controle.



Implementação paralela

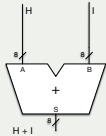
- Implementar um hardware que faça a seguinte operação:

$$Y = H + I + J + K + L - N - O$$

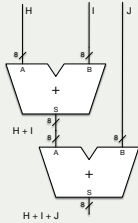
- Considere todas as variáveis de 8 bits.



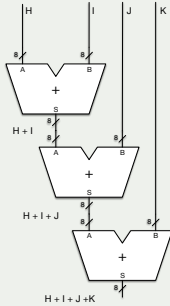
Implementando: $Y = H + I + J + K + L - N - O$



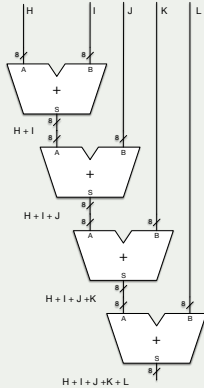
Implementando: $Y = H + I + J + K + L - N - O$



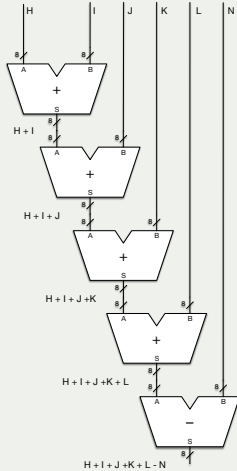
Implementando: $Y = H + I + J + K + L - N - O$



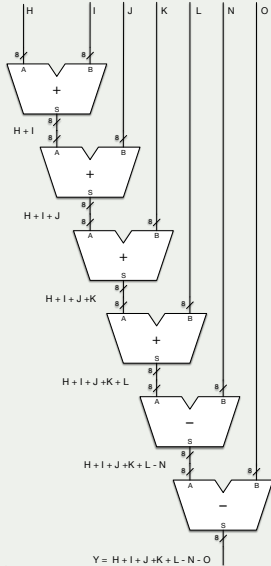
Implementando: $Y = H + I + J + K + L - N - O$



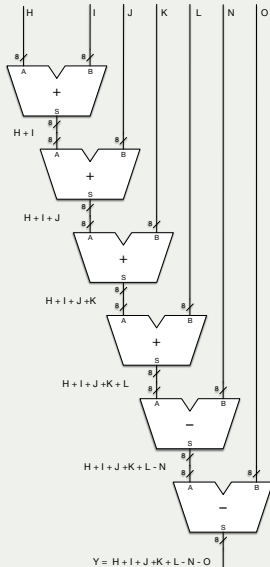
Implementando: $Y = H + I + J + K + L - N - O$



Implementando: $Y = H + I + J + K + L - N - O$



Implementando: $Y = H + I + J + K + L - N - O$



- Rápido, mas grande consumo de lógica.
- Modificação da expressão implica em modificação do hardware.
- Entrada de todos os operandos de uma vez.



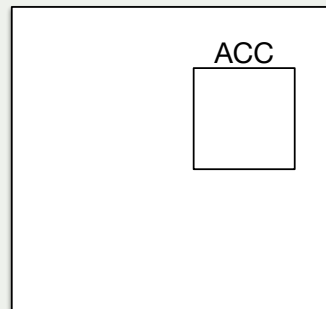
Como melhorar?

- Reutilizar ao máximo o hardware.
- Executar uma operação por vez.
- Exemplo: Calculadora simples.



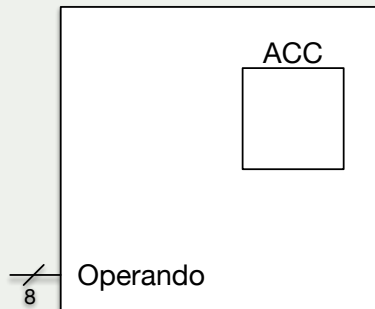
Como melhorar?

- Reutilizar ao máximo o hardware.
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- Exemplo: Calculadora simples.
- Comandos:



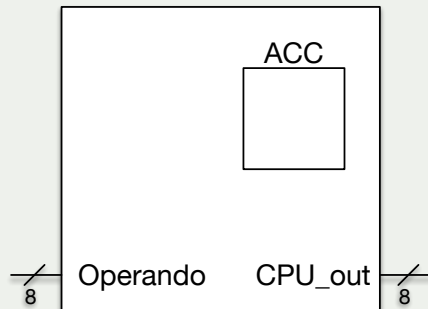
Como melhorar?

- Reutilizar ao máximo o hardware.
- Executar uma operação por vez.
- Exemplo: Calculadora simples.
- Comandos:
 - $ACC \leftarrow Operando$



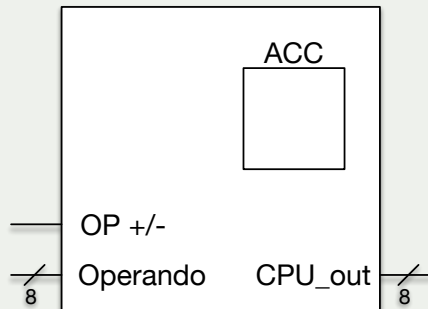
Como melhorar?

- Reutilizar ao máximo o hardware.
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- Exemplo: Calculadora simples.
- Comandos:
 - $ACC \leftarrow Operando$
 - $CPU_out \leftarrow ACC$



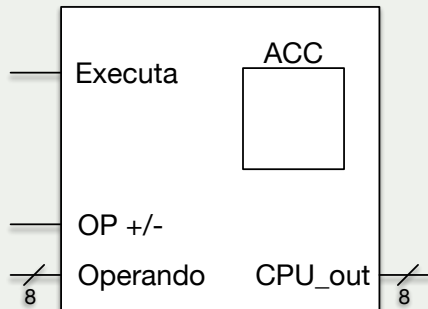
Como melhorar?

- Reutilizar ao máximo o hardware.
- Executar uma operação por vez.
- Exemplo: Calculadora simples.
- Comandos:
 - $ACC \leftarrow Operando$
 - $CPU_out \leftarrow ACC$
 - $ACC \leftarrow ACC + Operando$
 - $ACC \leftarrow ACC - Operando$



Como melhorar?

- Reutilizar ao máximo o hardware.
- Executar uma operação por vez.
- Exemplo: Calculadora simples.
- Comandos:
 - $ACC \leftarrow Operando$
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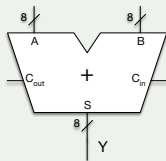


Rudi: Caminho de dados e sinais de controle

- Apenas um somador para soma e subtração:

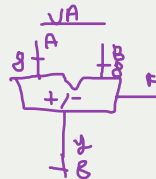
$$A - B = A + \bar{B} + 1$$

$$-B = \bar{B} + 1 \text{ complemento de 2}$$

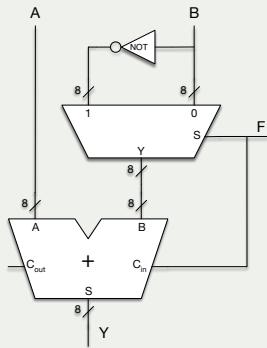


$$\begin{array}{r|l} F & y \\ \hline 0 & A + B + 0 = A + B \\ 1 & A + \bar{B} + 1 = A - B \end{array}$$

minha unidade aritmetica



- Apenas um somador para soma e subtração:

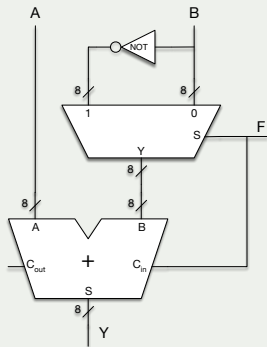


- $F = 0 \rightarrow Y = A + B$

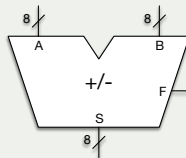
- $F = 1 \rightarrow Y = A + \overline{B} + 1 = A - B$



- Apenas um somador para soma e subtração:



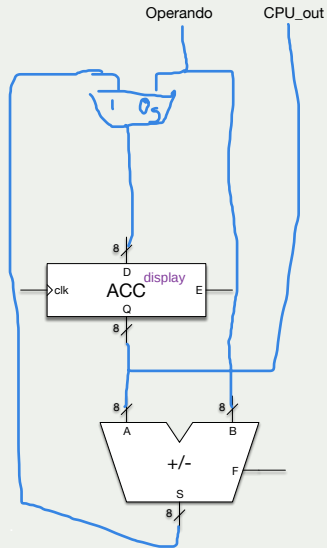
- Símbolo:



- $F = 0 \rightarrow Y = A + B$

- $F = 1 \rightarrow Y = A + \overline{B} + 1 = A - B$

Caminho dos dados

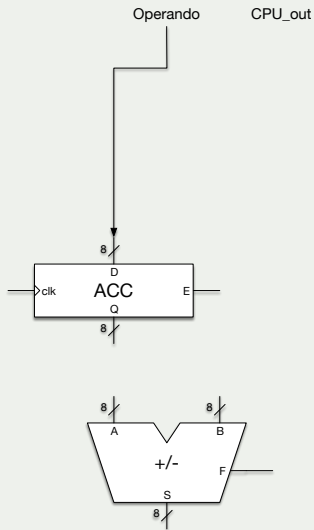


- $ACC \leftarrow Operando$
- $ACC \leftarrow ACC + Operando$
- $ACC \leftarrow ACC - Operando$
- $CPU_out \leftarrow ACC$

no display aparece é o registrador, uma memória, é o q tá dentro de um registrador



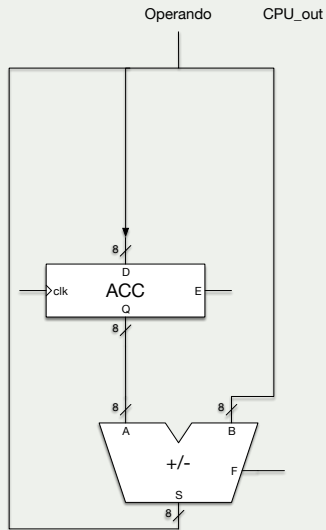
Caminho dos dados



- $ACC \leftarrow Operando$
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- $CPU_out \leftarrow ACC$



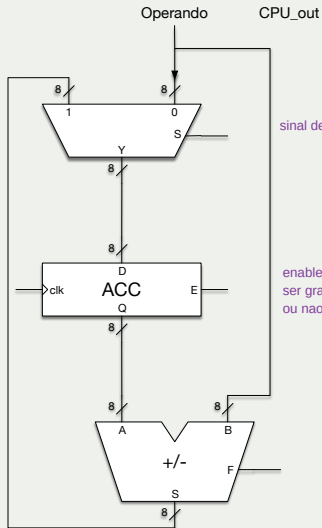
Caminho dos dados



- $ACC \leftarrow Operando$
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- $CPU_out \leftarrow ACC$



Caminho dos dados



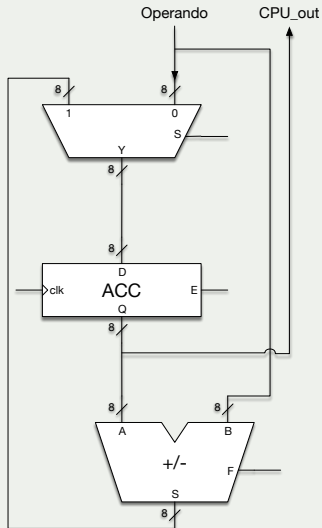
sinal de controle do s designa oq vai entrar no operador

enable define se vai
ser gravado no acumulador
ou nao

- $ACC \leftarrow Operando$
- $ACC \leftarrow ACC + Operando$
- $ACC \leftarrow ACC - Operando$
- $CPU_out \leftarrow ACC$



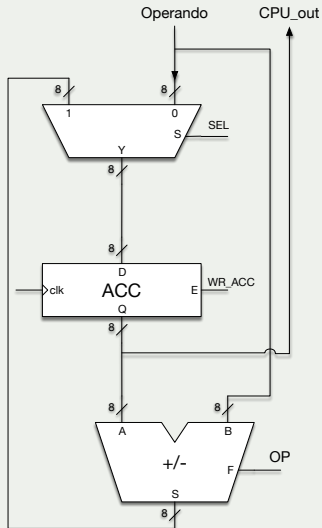
Caminho dos dados



- $ACC \leftarrow Operando$
- $ACC \leftarrow ACC + Operando$
- $ACC \leftarrow ACC - Operando$
- $CPU_out \leftarrow ACC$



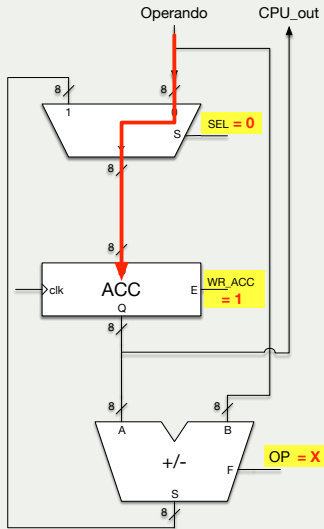
Caminho dos dados



- Uma entrada: Operando
- Uma saída: CPU_out
- Três sinais de controle internos:
 - SEL, WR_ACC e OP

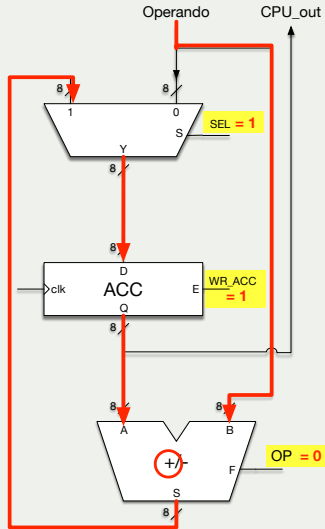


Caminho dos dados



■ $ACC \leftarrow Operando$

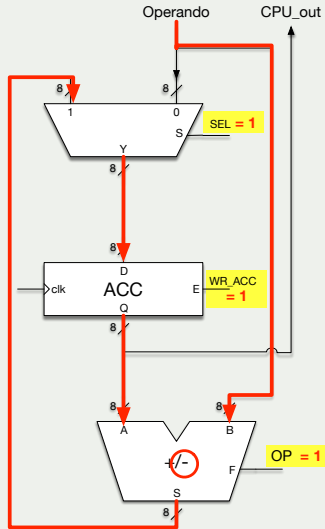
Caminho dos dados



$$ACC \leftarrow ACC + Operando$$



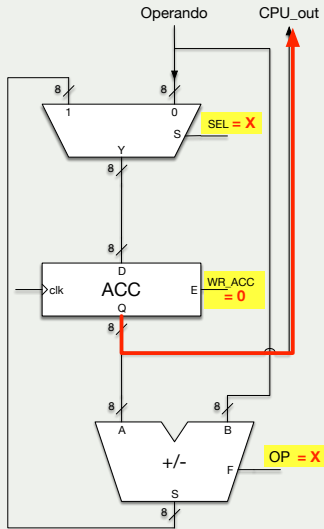
Caminho dos dados



■ $ACC \leftarrow ACC - Operando$



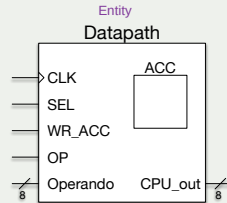
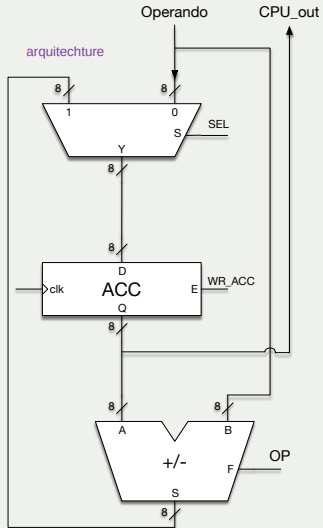
Caminho dos dados



■ $CPU_out \leftarrow ACC$



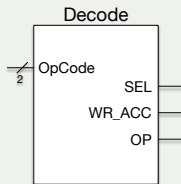
Caminho dos dados



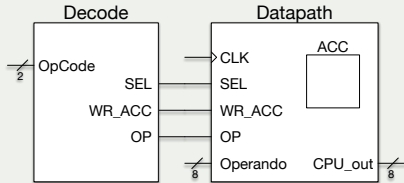
Operação	OpCode	SEL	WR_ACC	OP
$CPU_{out} \leftarrow ACC$	00	X	0	X
$ACC \leftarrow Operando$	01	0	1	X
$ACC \leftarrow ACC + Operando$	10	1	1	0
$ACC \leftarrow ACC - Operando$	11	1	1	1



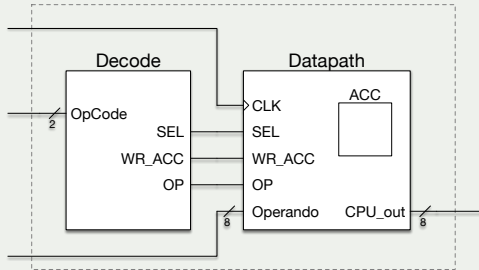
Operação	OpCode	SEL	WR_ACC	OP
$CPU_out \leftarrow ACC$	00	X	0	X
$ACC \leftarrow Operando$	01	0	1	X
$ACC \leftarrow ACC + Operando$	10	1	1	0
$ACC \leftarrow ACC - Operando$	11	1	1	1



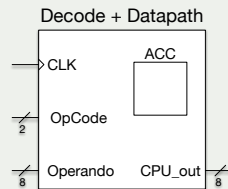
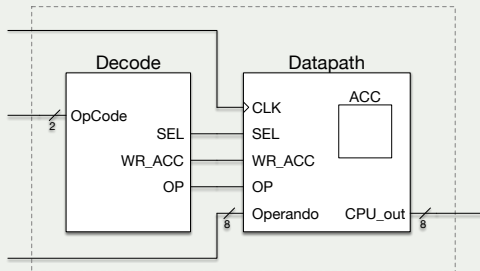
Caminho de dados e decodificador



Caminho de dados e decodificador



Caminho de dados e decodificador



- $H + I + J + K + L - N - O$

- $5 + 3 + 7 + 8 + 16 - 2 - 1$



■ $H + I + J + K + L - N - O$

■ $5 + 3 + 7 + 8 + 16 - 2 - 1$

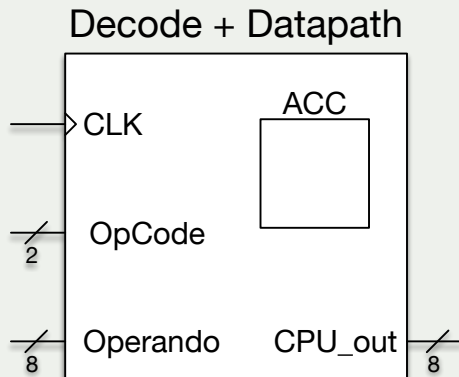
Operação	OpCode	Operando
$ACC \leftarrow 5$	01	00000101
$ACC \leftarrow ACC + 3$	10	00000011
$ACC \leftarrow ACC + 7$	10	00000111
$ACC \leftarrow ACC + 8$	10	00001000
$ACC \leftarrow ACC + 16$	10	00010000
$ACC \leftarrow ACC - 2$	11	00000010
$ACC \leftarrow ACC - 1$	11	00000001
$CPU_out \leftarrow ACC$	00	XXXXXXXX



■ $H + I + J + K + L - N - 0$

■ $5 + 3 + 7 + 8 + 16 - 2 - 1$

Operação	OpCode	Operando
$ACC \leftarrow 5$	01	00000101
$ACC \leftarrow ACC + 3$	10	00000011
$ACC \leftarrow ACC + 7$	10	00000111
$ACC \leftarrow ACC + 8$	10	00001000
$ACC \leftarrow ACC + 16$	10	00010000
$ACC \leftarrow ACC - 2$	11	00000010
$ACC \leftarrow ACC - 1$	11	00000001
$CPU_out \leftarrow ACC$	00	XXXXXXXX

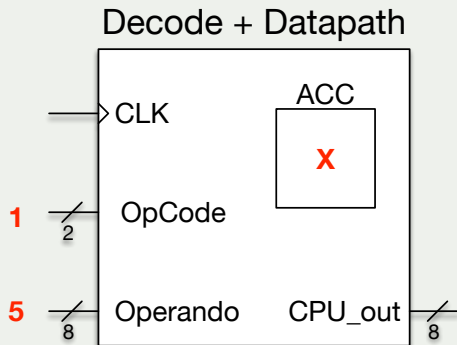


Calculando com o Rudi

■ $H + I + J + K + L - N - O$

■ $5 + 3 + 7 + 8 + 16 - 2 - 1$

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$ACC \leftarrow 5$	01	00000101
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$ACC \leftarrow ACC - 1$	11	00000001
$CPU_out \leftarrow ACC$	00	xxxxxxxx

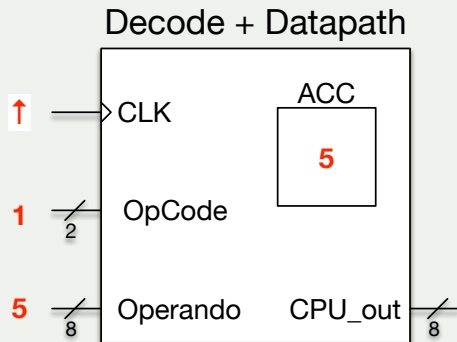


Calculando com o Rudi

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$ACC \leftarrow ACC - 1$	11	00000001
$CPU_out \leftarrow ACC$	00	xxxxxxxx

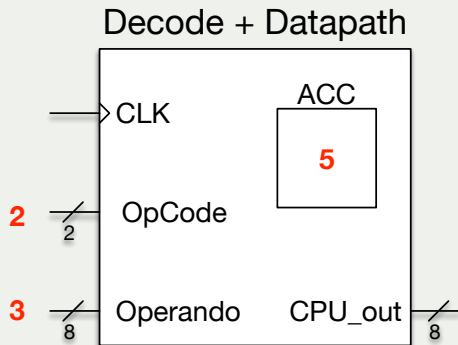


Calculando com o Rudi

■ $H + I + J + K + L - N - 0$

■ $5 + 3 + 7 + 8 + 16 - 2 - 1$

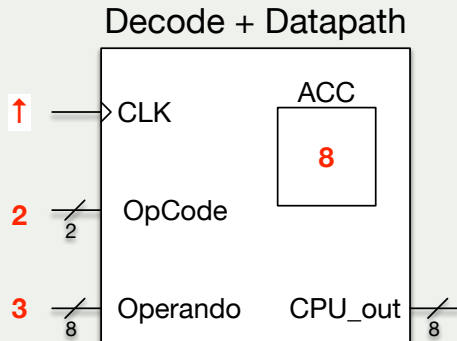
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$ACC \leftarrow ACC + 16$	10	00010000
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$CPU_out \leftarrow ACC$	00	XXXXXXXX



■ $H + I + J + K + L - N - 0$

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$ACC \leftarrow ACC - 1$	11	00000001
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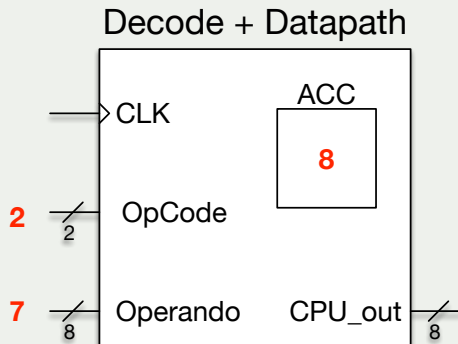


Calculando com o Rudi

■ $H + I + J + K + L - N - O$

■ $5 + 3 + 7 + 8 + 16 - 2 - 1$

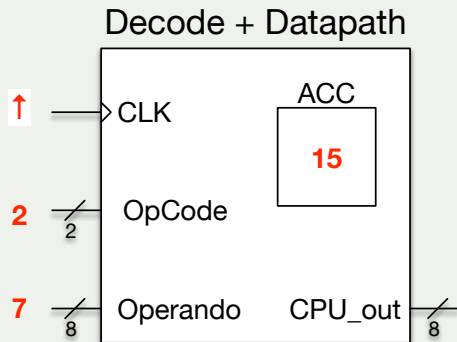
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$ACC \leftarrow ACC - 2$	11	00000010
$ACC \leftarrow ACC - 1$	11	00000001
$CPU_out \leftarrow ACC$	00	xxxxxxxx

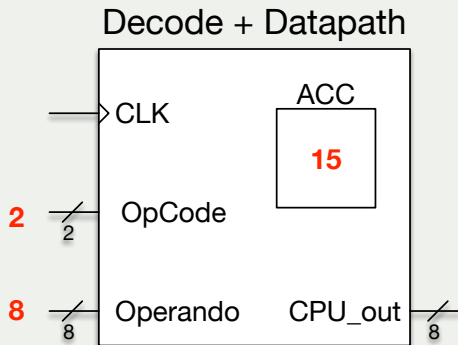


Calculando com o Rudi

$$\blacksquare H + I + J + K + L - N - O$$

$$\blacksquare 5 + 3 + 7 + 8 + 16 - 2 - 1$$

Operação	OpCode	Operando
$ACC \leftarrow 5$	01	00000101
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$ACC \leftarrow ACC - 2$	11	00000010
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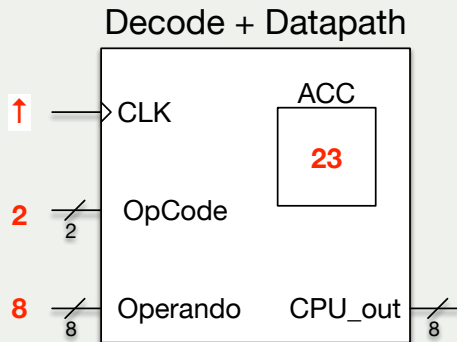


Calculando com o Rudi

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■ $5 + 3 + 7 + 8 + 16 - 2 - 1$

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$ACC \leftarrow ACC + 16$	10	00010000
$ACC \leftarrow ACC - 2$	11	00000010
$ACC \leftarrow ACC - 1$	11	00000001
$CPU_out \leftarrow ACC$	00	xxxxxxxx

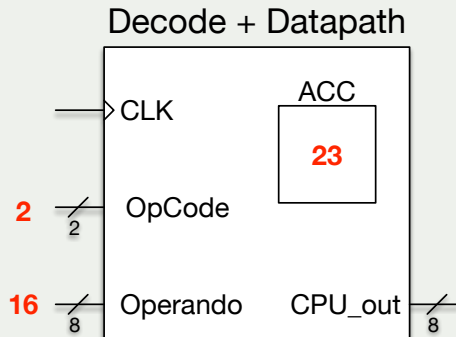


Calculando com o Rudi

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$CPU_out \leftarrow ACC$	00	XXXXXXXX

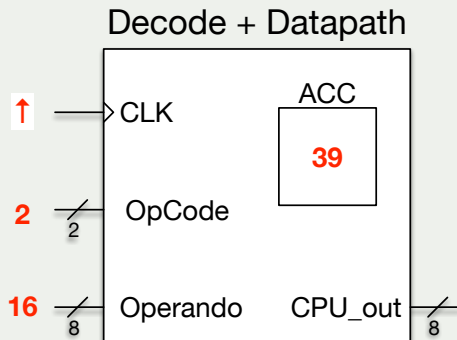


Calculando com o Rudi

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$CPU_out \leftarrow ACC$	00	XXXXXXXX

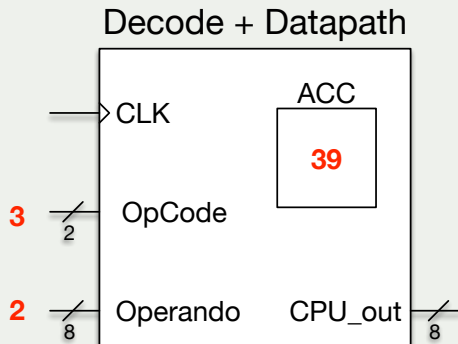


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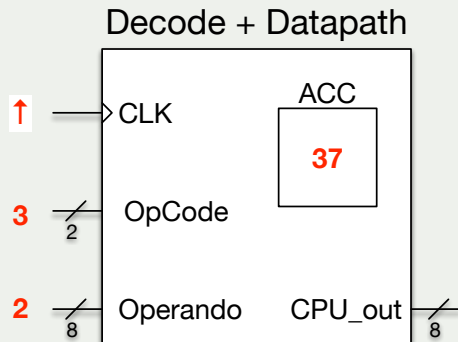
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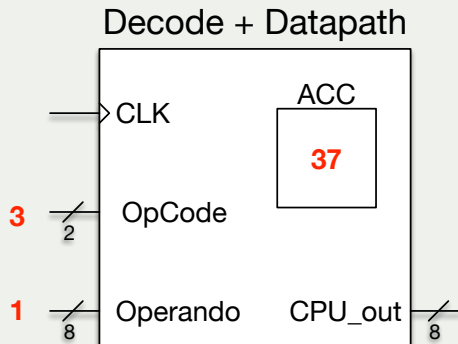


Calculando com o Rudi

■ $H + I + J + K + L - N - 0$

■ $5 + 3 + 7 + 8 + 16 - 2 - 1$

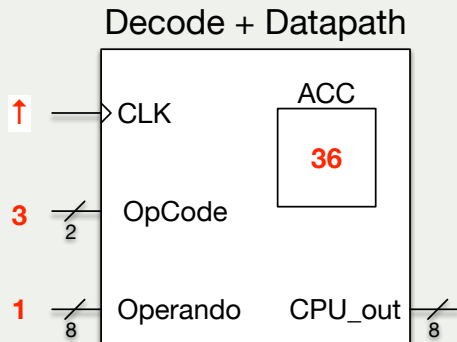
Operação	OpCode	Operando
$ACC \leftarrow 5$	01	00000101
$ACC \leftarrow ACC + 3$	10	00000011
$ACC \leftarrow ACC + 7$	10	00000111
$ACC \leftarrow ACC + 8$	10	00001000
$ACC \leftarrow ACC + 16$	10	00010000
$ACC \leftarrow ACC - 2$	11	00000010
$ACC \leftarrow ACC - 1$	11	00000001
$CPU_out \leftarrow ACC$	00	xxxxxxxx



■ $H + I + J + K + L - N - O$

■ $5 + 3 + 7 + 8 + 16 - 2 - 1$

Operação	OpCode	Operando
$ACC \leftarrow 5$	01	00000101
$ACC \leftarrow ACC + 3$	10	00000011
$ACC \leftarrow ACC + 7$	10	00000111
$ACC \leftarrow ACC + 8$	10	00001000
$ACC \leftarrow ACC + 16$	10	00010000
$ACC \leftarrow ACC - 2$	11	00000010
$ACC \leftarrow ACC - 1$	11	00000001
$CPU_out \leftarrow ACC$	00	xxxxxxxx

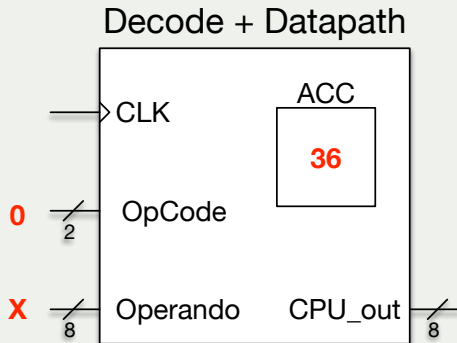


Calculando com o Rudi

■ $H + I + J + K + L - N - 0$

■ $5 + 3 + 7 + 8 + 16 - 2 - 1$

Operação	OpCode	Operando
$ACC \leftarrow 5$	01	00000101
$ACC \leftarrow ACC + 3$	10	00000011
$ACC \leftarrow ACC + 7$	10	00000111
$ACC \leftarrow ACC + 8$	10	00001000
$ACC \leftarrow ACC + 16$	10	00010000
$ACC \leftarrow ACC - 2$	11	00000010
$ACC \leftarrow ACC - 1$	11	00000001
$CPU_out \leftarrow ACC$	00	XXXXXXXX

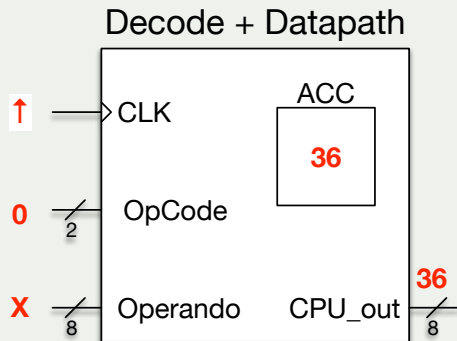


Calculando com o Rudi

■ $H + I + J + K + L - N - 0$

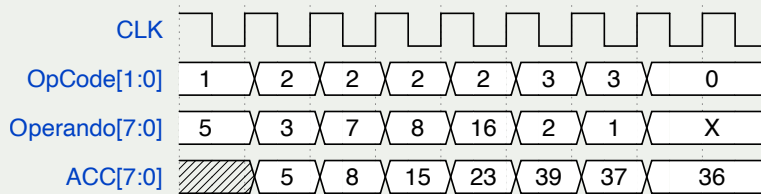
■ $5 + 3 + 7 + 8 + 16 - 2 - 1$

Operação	OpCode	Operando
$ACC \leftarrow 5$	01	00000101
$ACC \leftarrow ACC + 3$	10	00000011
$ACC \leftarrow ACC + 7$	10	00000111
$ACC \leftarrow ACC + 8$	10	00001000
$ACC \leftarrow ACC + 16$	10	00010000
$ACC \leftarrow ACC - 2$	11	00000010
$ACC \leftarrow ACC - 1$	11	00000001
$CPU_out \leftarrow ACC$	00	XXXXXXXX



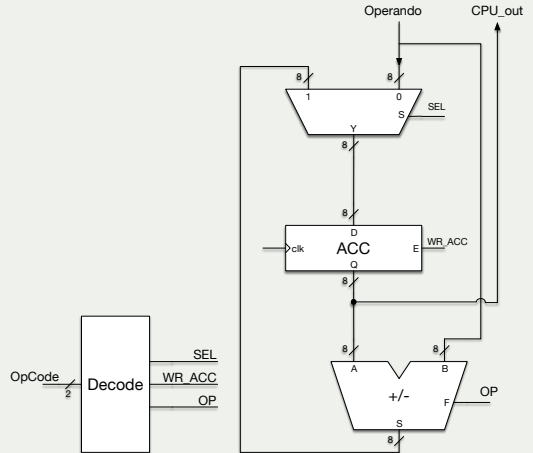
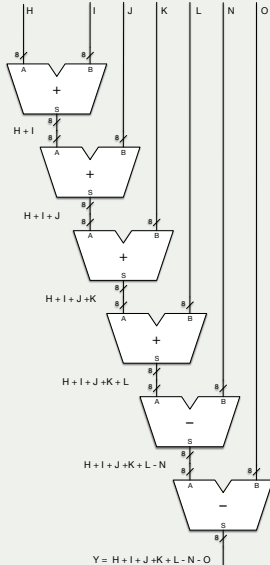
Calculando com o Rudi – Forma de onda

Operação	OpCode	Operando
$ACC \leftarrow 5$	01	00000101
$ACC \leftarrow ACC + 3$	10	00000011
$ACC \leftarrow ACC + 7$	10	00000111
$ACC \leftarrow ACC + 8$	10	00001000
$ACC \leftarrow ACC + 16$	10	00010000
$ACC \leftarrow ACC - 2$	11	00000010
$ACC \leftarrow ACC - 1$	11	00000001
$CPU_out \leftarrow ACC$	00	XXXXXXXX



Comparação

Implementação Paralela vs. Rudi



■ Implementação paralela:

- 4 somadores e 2 subtratores
- Entrada dos operandos em paralelo (pode ser mais rápida)
- Fixa, apenas para a expressão projetada

■ Rudi:

- 1 somador, 2 multiplexadores, 1 decodificado e 1 registrador
- Entrada dos operandos sequenciais
- Flexível, pode implementar qualquer expressão com soma e subtração



FIM!